

A Literature-Implied Negative Answer to the Critical-Exponent Frame-Potential Conjecture

Status note for arXiv:1504.02023

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Status. `literature_implied_answer` (`full negative answer`). The decisive later classification is known; the short implication below is included to match it precisely to the older conjecture. It is not claimed as a new counterexample.

Source question

Kasso A. Okoudjou, *Preconditioning techniques in frame theory and probabilistic frames*, arXiv:1504.02023, Section 3.2, PDF page 29, uses

$$\text{FP}_{p,M}(\Phi) = \sum_{k,\ell=1}^M |\langle \varphi_k, \varphi_\ell \rangle|^p$$

for collections of unit vectors. For $N \geq 3$ it records the conjecture that, with

$$q_N := \frac{\log(N(N+1)/2)}{\log N},$$

every $\Phi \subset S^{N-1}$ of cardinality $N+1$ satisfies

$$\text{FP}_{q_N, N+1}(\Phi) \geq N+3,$$

with equality only for an orthonormal basis plus one repeated vector or an equiangular finite unit-norm tight frame.

Supporting theorem

Zhiqiang Xu and Zili Xu, *The minimizers of the p -frame potential*, arXiv:1907.10861, define the lifted ETF L_k^d in equation (6), PDF page 3. Their Theorem 1.2, PDF pages 3–4, completely determines the minimizers among $d+1$ unit vectors in $\mathbb{R}^d$ for every $0 < p < 2$. Their convention sums only over $i \neq j$, which differs from Okoudjou’s convention by the constant $d+1$.

The theorem answers a later lifted-ETF phase-diagram conjecture and does not explicitly label the exact 2015 sentence above as a target. The following identification extracts its direct consequence for that sentence.

Identification and strict counterexample

Fix $N \geq 3$. In the first two coordinates of $\mathbb{R}^N$ set

$$u_1 = (1, 0), \quad u_2 = \left(-\frac{1}{2}, \frac{\sqrt{3}}{2}\right), \quad u_3 = \left(-\frac{1}{2}, -\frac{\sqrt{3}}{2}\right),$$

and adjoin $e_3, \dots, e_N$. This is the lifted ETF L_2^N of Xu–Xu: a planar Mercedes triple plus an orthonormal basis of its orthogonal complement. The only nonzero off-diagonal inner products are the six ordered pairs among u_1, u_2, u_3 , each of absolute value $1/2$. Consequently

$$\text{FP}_{q_N, N+1}(L_2^N) = N + 1 + 6 \cdot 2^{-q_N}.$$

It remains only to compare exponents. Put $a = \log_2(3/2)$. For $h(t) = (t+1)/(2t^a)$, one has $h(2) = 1$ and

$$\text{sign } h'(t) = \text{sign}((1-a)t - a).$$

Since $a < 2/3$, the latter expression is positive for $t \geq 2$. Hence $(N+1)/2 > N^a$ for every integer $N \geq 3$. Taking logarithms base N gives

$$q_N = 1 + \log_N \left(\frac{N+1}{2} \right) > 1 + a = \log_2 3.$$

Therefore $6 \cdot 2^{-q_N} < 2$, and so

$$\boxed{\text{FP}_{q_N, N+1}(L_2^N) < N + 3}$$

for every $N \geq 3$. The conjecture is false in every dimension in which it was posed.

Provenance and scope

The later paper proves more than this strict counterexample: it classifies the global minimizers throughout $0 < p < 2$. The present status note contributes only the direct identification of its known lifted configuration with the older critical exponent. No part of the source conjecture remains open.

References

- [1] K. A. Okoudjou, *Preconditioning techniques in frame theory and probabilistic frames*, arXiv:1504.02023 (2015), Section 3.2, p. 29.
- [2] Z. Xu and Z. Xu, *The minimizers of the p -frame potential*, arXiv:1907.10861 (2019; revised 2020), Definition (6) and Theorem 1.2, pp. 3–4.